

# Frontend Specification Document

## Swasthya Katha — Magazine Reader and Accessible Learning Interface

**Version:** 1.0

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### 1. Purpose

This document defines the user interface, interaction behaviour, responsive layout, reader modes, motion, accessibility, states, and frontend quality requirements.

### 2. Frontend goals

The frontend must:

- Feel like a premium magazine.
- Remain understandable for users with limited literacy.
- Work without WebGL.
- Work without audio.
- Work in a linear reader.
- Support local languages.
- Perform on low-end Android devices.
- Preserve privacy.
- Make the next safe action clear.

### 3. Experience modes

#### 3.1 Magazine mode

Use for immersive editorial reading:

- Two-page desktop spread.
- Single-page mobile layout.
- Page-turn animation.
- Paper-like materials.
- Chapter tabs.
- Visual spreads.
- Optional enhanced physics.

## 3.2 Linear mode

Use for accessibility, low-end devices, and users who prefer simple reading:

- Semantic document flow.
- One lesson block after another.
- No required page flip.
- Screen-reader-friendly headings.
- Transcript and visual descriptions.
- Stable deep links.

## 3.3 Facilitator mode

Use for health workers and group teaching:

- Large scene display.
- Presenter controls.
- Discussion pauses.
- High-contrast projection.
- Read-aloud and captions.
- Optional lesson navigation without progress tracking.

## 4. Information hierarchy

Every screen must answer:

1. What is this?
2. What is the one important message?
3. What does the visual show?
4. What should I do next?
5. Where can I get help?

## 5. Route-level requirements

```
/[locale]
/[locale]/reader/[publication]
/[locale]/reader/[publication]/[chapter]
/[locale]/learn/[lesson]
/[locale]/help
/[locale]/privacy
/[locale]/accessibility
```

Every route must:

- Set the correct document language.

- Provide translated metadata.
- Support direct loading.
- Provide a useful loading state.
- Provide an error state.
- Work under reduced motion.

## 6. Screen specifications

### 6.1 Home

Elements:

- Publication title.
- One-sentence purpose.
- Language selector.
- Start reading button.
- Listen button.
- Simple mode button.
- Privacy link.
- Help link.

Acceptance:

- New user can identify the start action without assistance.
- Language selector is keyboard and screen-reader accessible.
- No sensitive detail is exposed in unnecessary preview text.

### 6.2 Cover

Elements:

- Title.
- Illustration.
- Edition/update information.
- Start button.
- Reading mode selector.
- Optional content notice.

Acceptance:

- Cover opens with a stable fallback if animation fails.
- Start is available without precise gesture.

## 6.3 Lesson

Elements:

- Chapter label.
- Lesson title.
- Key message.
- Visual scene or fallback.
- Narration controller.
- Transcript.
- Action card.
- Progress.
- Next/previous controls.

Acceptance:

- Main message is visible without scrolling excessively.
- Text remains readable with increased size.
- Visual does not hide the action.

## 6.4 Help

Elements:

- Product limitations.
- Professional-care guidance.
- Referral information.
- Privacy guidance.
- Emergency wording where approved.

## 7. Reader interaction

### Basic navigation

- Next button.
- Previous button.
- Table of contents.
- Chapter selector.
- Keyboard arrows where appropriate.
- Swipe on touch devices.
- Direct lesson links.

## Page flip

Stages:

```
idle
  -> pressed
  -> dragging
  -> commit
  -> completing
  -> settled
```

Rules:

- Dragging must not trap the user in the book.
- Release below threshold returns the page smoothly.
- Successful flip updates state once.
- Focus moves to the new page heading.
- Screen reader receives an announcement.

## 8. Realistic page requirements

The production effect may include:

- Spine-origin rotation.
- Page depth.
- Dynamic shadow.
- Paper grain.
- Page curl approximation.
- Corner drag.
- Velocity-based commit.
- Subtle crease lighting.

The effect must not:

- Carry essential information only in a canvas.
- Prevent scrolling.
- Require a mouse.
- Break reduced motion.
- Block the page from loading.

## 9. Visual scene requirements

Each scene must have:

- Scene label.
- Full description.
- Target registry.
- Text equivalent.
- Static fallback.
- Loading state.
- Error state.
- Focus state.
- Reduced-motion state.

Target interaction:

```
Scene loaded
-> target list available
-> narration selects target
-> target highlights
-> user can manually select target
-> description appears
```

## 10. Narration interface

Controls:

- Play/pause.
- Seek.
- Speed.
- Replay.
- Volume.
- Transcript.
- Language.

Requirements:

- No autoplay by default.
- Text remains available.
- Audio pause pauses synchronized motion.
- Transcript is keyboard accessible.
- Audio failure produces a readable fallback.

## 11. Highlighting behaviour

Text highlighting:

- Current word receives a clear but non-flashing highlight.
- Previous words remain readable.
- Highlight colour is not the only cue.
- Highlighting stops or simplifies under reduced motion.

Visual highlighting:

- Target receives glow, outline, scale, or trace.
- Unrelated content may dim, but must remain discoverable.
- Manual target selection is supported.
- Target descriptions are accessible.

## 12. Responsive rules

### Mobile

- Single-page view.
- Bottom-reachable controls.
- Minimal decoration.
- Swipe plus buttons.
- Large type.
- Vertical scene fallback.

### Tablet

- Single or two-page view by orientation.
- Larger visual canvas.
- Optional chapter rail.

### Desktop

- Two-page spread.
- Chapter navigation.
- Enhanced effects.
- Keyboard shortcuts.
- Facilitator layout.

### 13. Typography requirements

- Body text must remain comfortable at user-selected size.
- No essential copy inside raster images.
- Test every launch script.
- Avoid overly thin weights.
- Use line height suitable for new readers.
- Prevent clipping and overflow after translation expansion.

### 14. Design states

Every component needs:

- Default.
- Hover.
- Focus.
- Pressed.
- Disabled.
- Loading.
- Offline.
- Error.
- Success.

### 15. Motion specification

- Motion explains relationships or transitions.
- Background motion is minimal.
- No flashing.
- No forced parallax.
- Animation pauses during audio pause where synchronized.
- Reduced-motion users receive static or simplified transitions.

The browser's `prefers-reduced-motion` setting should be honoured. [web:23][web:24]

### 16. Error states

#### Publication unavailable

Display a simple message, retry action, and available offline content if present.

## **Audio unavailable**

Display transcript and readable text.

## **Visual unavailable**

Display scene label, explanation, and fallback asset.

## **Translation incomplete**

Display the fallback language clearly and never silently mix critical instructions.

## **Offline**

Display whether the current content is cached and whether content may be stale.

## **17. Accessibility requirements**

- Semantic landmarks.
- Heading hierarchy.
- Skip link.
- Keyboard operation.
- Focus restoration.
- Accessible names.
- Text alternatives.
- Captions/transcripts.
- Contrast.
- Text scaling.
- Reduced motion.
- No colour-only meaning.
- Complete linear mode.

## **18. Frontend acceptance checklist**

- [ ] Home start action is clear.
- [ ] Locale switch works.
- [ ] Magazine mode works.
- [ ] Linear mode works.
- [ ] Keyboard navigation works.
- [ ] Touch navigation works.
- [ ] Focus changes are announced.
- [ ] Narration works.

- ☐ Transcript works.
- ☐ Visual target works.
- ☐ Static fallback works.
- ☐ Reduced motion works.
- ☐ Text scaling works.
- ☐ Error states work.
- ☐ Offline state is understandable.
- ☐ Low-end device test passes.